

Nmap & Network Recon – Quick Reference

The scanning, discovery and enumeration commands that come up on almost every engagement, grouped by what you're trying to learn about a target. Swap the example IPs/hosts for your actual scope.

1. Host Discovery

nmap -sn — ping scan — find live hosts, don't scan ports `nmap -sn 10.10.10.0/24`

nmap -PR — ARP ping scan — fast, but local subnet only `nmap -PR 10.10.10.0/24`

nmap -PS — TCP SYN ping on specific ports `nmap -PS22,80,443 10.10.10.0/24`

nmap -PU — UDP ping on specific ports `nmap -PU53,161 10.10.10.0/24`

nmap -PE — ICMP echo ping `nmap -PE 10.10.10.0/24`

nmap -sL — list targets and resolve DNS, don't scan at all `nmap -sL 10.10.10.0/24`

nmap -n — skip DNS resolution — noticeably faster on big ranges `nmap -n -sn 10.10.10.0/24`

2. Port Scanning

nmap -sS — TCP SYN scan — the default, needs root, doesn't complete the handshake `nmap -sS 10.10.10.5`

nmap -sT — TCP connect scan — full handshake, works without root `nmap -sT 10.10.10.5`

nmap -sU — UDP scan — slower, but DNS/SNMP/DHCP live here `nmap -sU 10.10.10.5`

nmap -sA — ACK scan — maps firewall rules, not open ports `nmap -sA 10.10.10.5`

nmap -p- — scan all 65535 TCP ports `nmap -p- 10.10.10.5`

nmap -p — scan a specific list of ports `nmap -p 22,80,443 10.10.10.5`

nmap -p1-1000 — scan a port range `nmap -p1-1000 10.10.10.5`

nmap -F — fast scan — top 100 most common ports only `nmap -F 10.10.10.5`

nmap --top-ports — scan the N most common ports `nmap --top-ports 20 10.10.10.5`

3. Service, Version & OS Detection

nmap -sV — probe open ports to identify service/version `nmap -sV 10.10.10.5`

nmap -sV --version-intensity — more thorough (and slower) version probing, 0-9
`nmap -sV --version-intensity 9 10.10.10.5`

nmap -sV --version-light — faster, lighter version probing `nmap -sV --version-light 10.10.10.5`

nmap -O — remote OS fingerprinting `nmap -O 10.10.10.5`

nmap -O --osscan-guess — guess harder on ambiguous OS matches `nmap -O --osscan-guess 10.10.10.5`

nmap -A — aggressive: OS + version + default scripts + traceroute `nmap -A 10.10.10.5`

nmap --banner (via nc) — manual banner grab when -sV isn't enough `nc -nv 10.10.10.5 21`

4. NSE Scripts

nmap -sC — run the default set of safe NSE scripts

```
nmap -sC 10.10.10.5
```

nmap --script=vuln — run vulnerability-detection scripts

```
nmap --script=vuln 10.10.10.5
```

nmap --script=safe — run only scripts flagged safe for production targets

```
nmap --script=safe 10.10.10.5
```

nmap --script=http-enum — enumerate common web paths/directories

```
nmap --script=http-enum -p80 10.10.10.5
```

nmap --script=smb-enum-shares — enumerate SMB shares

```
nmap --script=smb-enum-shares -p445 10.10.10.5
```

nmap --script=ssl-enum-ciphers — enumerate supported TLS ciphers

```
nmap --script=ssl-enum-ciphers -p443 10.10.10.5
```

nmap --script-help — show usage/args for a specific script

```
nmap --script-help=http-enum
```

nmap --script-updatedb — refresh the local NSE script database

```
nmap --script-updatedb
```

5. Timing & Performance

nmap -T0 ... -T5 — timing templates — 0 paranoid, 5 insane

```
nmap -T4 10.10.10.5
```

nmap -T4 — aggressive timing — the sane default for most scans

```
nmap -T4 10.10.10.5
```

nmap --min-rate — send at least N packets per second

```
nmap --min-rate 1000 10.10.10.5
```

nmap --max-retries — cap retransmission attempts on unresponsive ports

```
nmap --max-retries 1 10.10.10.5
```

nmap --host-timeout — abandon a slow/unresponsive host entirely

```
nmap --host-timeout 30s 10.10.10.0/24
```

nmap --stats-every — print progress on a fixed interval

```
nmap --stats-every 10s 10.10.10.0/24
```

6. Firewall / IDS Evasion

nmap -f — fragment packets to slip past simple packet filters

```
nmap -f 10.10.10.5
```

nmap -D — decoy scan — hide your source among fake ones

```
nmap -D RND:10 10.10.10.5
```

nmap -S — spoof the source address

```
nmap -S 10.10.10.99 10.10.10.5
```

nmap --source-port — scan from a port likely to be trusted by a filter

```
nmap --source-port 53 10.10.10.5
```

nmap --data-length — pad packets to alter their signature

```
nmap --data-length 25 10.10.10.5
```

nmap --mtu — set a custom packet MTU to fragment differently

```
nmap --mtu 8 10.10.10.5
```

7. Output Formats

nmap -oN — normal text output, saved to a file `nmap -oN scan.txt 10.10.10.5`

nmap -oX — XML output — feeds into other tooling cleanly `nmap -oX scan.xml 10.10.10.5`

nmap -oG — grepable output — quick to pipe through grep/awk `nmap -oG scan.gnmap 10.10.10.5`

nmap -oA — write normal + XML + grepable all at once `nmap -oA full_scan 10.10.10.5`

nmap -v / -vv — increase output verbosity `nmap -vv 10.10.10.5`

8. Other Recon Tools

masscan — extremely fast full-port sweep across large ranges

`masscan -p1-65535 10.10.10.0/24 --rate=1000`

rustscan — fast port discovery, hands results straight to nmap `rustscan -a 10.10.10.5`

nc -zv — quick TCP port check with netcat `nc -zv 10.10.10.5 20-100`

nc <host> <port> — manual banner grabbing — connect and read what greets you `nc -nv 10.10.10.5 80`

whois — domain/IP registration and ownership lookup `whois example.com`

dig ANY — pull whatever DNS records are exposed `dig example.com ANY`

dig axfr — attempt a DNS zone transfer (often blocked, worth trying)

`dig axfr @ns1.example.com example.com`

nslookup — basic DNS lookup `nslookup example.com`

tracert — trace the network path packets take to a host `tracert 10.10.10.5`

9. Useful Combos

first pass — a solid all-around opening scan `nmap -sS -sV -O -T4 -oA full_scan 10.10.10.5`

two-phase scan — fast full sweep, then deep-dive only the open ports

`nmap -p- --min-rate 5000 -T4 10.10.10.5`

scan a list of targets — read hosts from a file instead of one at a time

`nmap -iL targets.txt -oA sweep`

targeted vuln check — run vuln scripts only against ports already known open

`nmap --script vuln -p22,80,443 10.10.10.5`

quick live-host sweep — find what's alive before you enumerate anything

`nmap -sn 10.10.10.0/24 -oG - | grep Up`

Scope reminder: Aggressive timing, ACK scans, decoys and evasion flags are the kind of thing that gets a SOC's attention fast. Only run these against scope you're actually authorised to test.